

HMFA MEMORIAL INSTITUTE OF ENGINEERING AND TECHNOLOGY



Affiliated By

Dr. APJ Abdul Kalam Technical University, Lucknow



Session 2023-26

Department of Electronic & Communication Engineering

Project Report

on

Bluetooth & Automatic Controlled Firefighter

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Session 2023-26

CERTIFICATE

This is to certify that the project entitled “**Bluetooth & Automatic Controlled Programmable Firefighter Robot**” submitted by **Ram Kishun, Shyam Kumar, Rupesh Kumar and Sumit Gupta** in fulfilment of the requirement for the award of degree of B-TECH in “**Department of Electronics and Communication Engineering**” session **2023-26** of “**Dr. A.P.J. Abdul Kalam Technical University**”, Uttar Pradesh is record of the candidate own work carried out by him under our provision the matter embodied in the this thesis. It is original and has not submitted for the award any other degree.

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DECLARATION

Project title: To Fabricate Bluetooth & Automatic Controlled Programmable Firefighter Robot

We **Ram Kishun, Shyam Kumar, Sumit Gupta and Rupesh Kumar of ECE Department 2023-2026** certify that this project is our own work, based on our personal study and research and that we have acknowledged all material and sources used in its preparation, whether they be books, articles, reports, lecture notes, and any other kind of document, electronic or personal communication.

We also certify that this project has not previously been submitted for assessment in any academic capacity, and that we have not copied in part or whole or otherwise plagiarised the work of other persons.

We confirm that we have identified and declared all possible conflicts that we may have.

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TABLE OF CONTENTS

Contents	Page No.
➤ ABSTRACT	5
➤ ACKNOWLEDGEMENT	6
➤ INTRODUCTION	7
➤ OBJECTIVE	8
➤ BLOCK DIAGRAM	8
➤ FLOW CHART	9
➤ CIRCUIT DIAGRAM	10
➤ TABLE OF REQUIRED COMPONENT'S & TOOLS.....	11
➤ ELECTRONIC COMPONENT'S	
➤ Arduino Uno	12
➤ Motor Driver	15
➤ LED	17
➤ BO Motor	17
➤ IR Flame Sensor	18
➤ Bluetooth HC-05	19
➤ Relay	23
➤ Servo Motor	25
➤ Li-Battery	27
➤ Resistance	28
➤ Battery Charging Module	29
➤ PCB	29
➤ Wheels	30
➤ MECHANICAL COMPONENT'S	
➤ Buzzer	31
➤ Water Pump	31
➤ Rocker Switch	32
➤ TOOLS	33
➤ WORKING: Hardware , Software, Coding	33
➤ ADVANTAGE	42
➤ DISADWANTAGE.....	42
➤ PROJECT COST	43
➤ FUTURE SCOPE	44
➤ APPLICATION	44
➤ RESULT	45
➤ CONCLUSION.....	46
➤ REFERENCE	47

ABSTRACT

There are many possibilities to get fire in an industry or any remote areas for examples in cotton mills, garments fuel storage etc.

Electric leakage causes huge damage also it, a worst -case scenario, which causes heavy loss not only surroundings got affected and very loss. Nowadays the fire Extinguisher are implements in the industry, which is mostly manual one or sensible annunciator. In our project we choose desired. Application which opens the door for modern and future world. Robotics is the emerging solution to protect the humans, their wealth and surroundings. The aim of our project is to design a firefighting Robot using embedded system.

We have chosen the ATMEGA 8 AVR Controller, which exactly having performance. and low power 8-bit controller, which exactly having the required features.

This project helps to improve the safety to us in our day -to-day life.

ACKNOWLEDGEMENT

This is major project part of final year project, title of the firefighting robot. We are really thankful to our Department of Electronics and Communication Engineering HOD Mam (Mrs. Neetu Singh), collage HMFA Memorial institute of Engineering and Technology for his invaluable guidance and assistance, without which the accomplishment of the task would have never been possible.

We are also thankful to HOD Mam (Mrs. Neetu Singh), our project mentor for giving this opportunity to explore into the real word and realise the interrelation without which a project can never progress. In our present we have chosen the topic “ **Bluetooth & Automatic Controlled Programmable Firefighter Robot**”.

We are also thankful to our present, friends and all staff of Electronics and Communication Engineering Department for providing us relevant information and necessary clarification support.

INTRODUCTION

- Aims to develop NODE MCU based firefighting robot.
- Monitors the areas where natural calamities and fire.
- Sense fire ,smoke and temperature at the site of disaster by using flame and smoke sensors mounted on the robot.
- If fire is detected with the help of sensor, MCU operators water pump mechanism.

Firefighting is a profession aimed at controlling and extinguishing fire. A person who engages in firefighting is known as a firefighter or fireman. Firefighters typically undergo a high degree of technical training. This involves structural firefighting and wildland firefighting. Specialized training includes aircraft firefighting, shipboard firefighting, aerial firefighting, maritime firefighting, and proximity firefighting.

Firefighting is a dangerous profession due to the toxic environment created by combustible materials. With the major risks being smoke, oxygen deficiency, elevated temperatures, poisonous atmospheres, and violent air flows.^[4] To combat some of these risks, firefighters carry self-contained breathing apparatus. Additional hazards include falls a constant peril while navigating unfamiliar layouts or confined spaces amid shifting debris under limited visibility and structural collapse that can exacerbate the problems encountered in a toxic environment.

The first step in a firefighting operation is reconnaissance to search for the location of the fire and to identify the specific risks. Fires can be extinguished by water, fuel or oxidant removal, or chemical flame inhibition; though, because fires are classified depending on the elements involved, such as grease, paper, electrical, etcetera, a specific type of fire extinguisher may be required. The classification is based on the type of fires that the extinguisher is more suitable for. In the United States, the types of fire are described by the National Fire Protection Association.

OBJECTIVE

- To detect fire in the disaster-prone area.
- Also provide audio and visual indications.
- Extinguishes fire on detection.
- Reduce the efforts of human labour and level of destruction.

BLOCK DIAGRAM :-

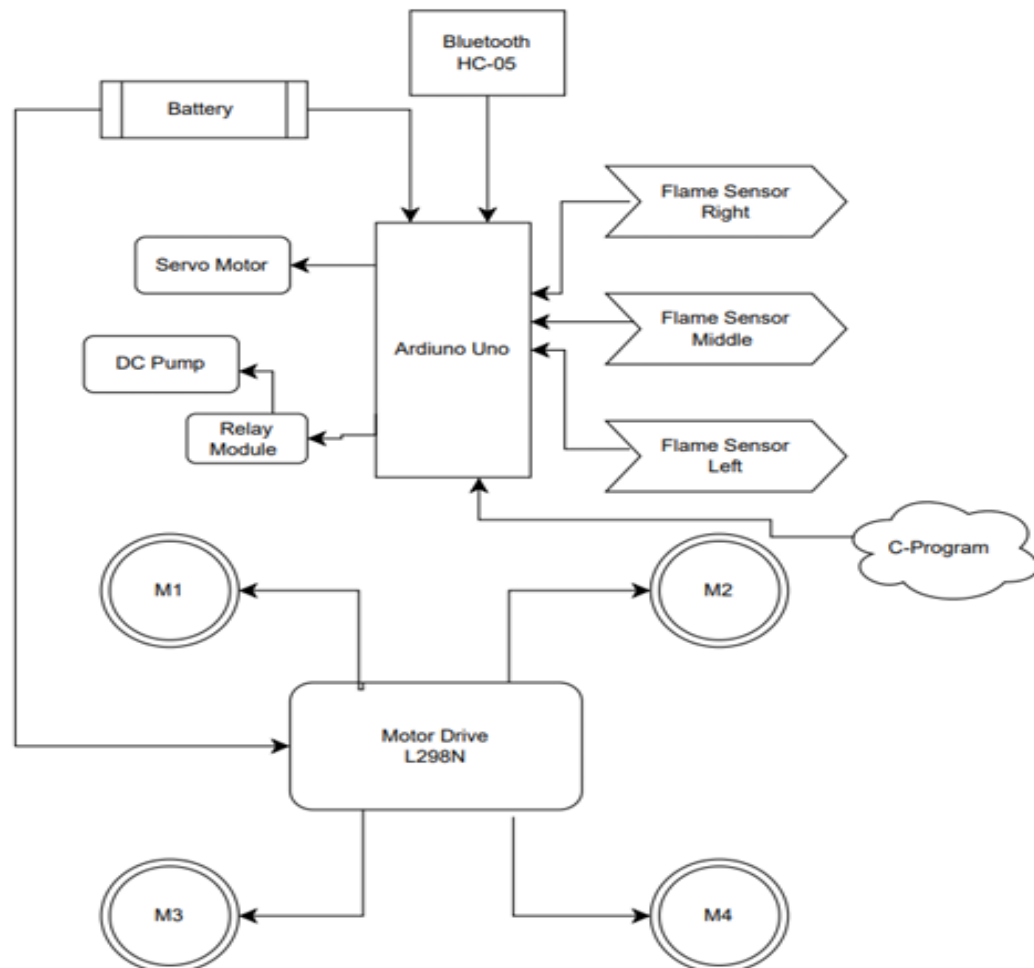


Fig 1. Block Diagram of Fire Fighting Robot

The block diagram illustrates the system architecture of the IoT-Based Fire Fighter Robot, highlighting the interaction between hardware components managed by an Arduino Uno microcontroller. The robot is powered by a rechargeable Li-ion battery that supplies energy to all components, ensuring untethered mobility. The core processing unit, Arduino Uno, acts as the central controller, interfacing with various modules for sensing, control, and communication. Three flame sensors positioned on the left, middle, and right of the robot—continuously monitor the surrounding environment for fire. When a flame is detected, the corresponding sensor transmits the signal to the Arduino, which then executes pre-programmed logic written in C. For movement, the Arduino interfaces with an L298N motor driver module, which controls four DC gear motors (M1 to M4), enabling the robot to navigate toward the detected fire autonomously. The robot's motion can also be overridden or directed manually via a Bluetooth HC-05 module, which communicates with a smartphone for remote operation. Once the flame location is identified, the Arduino activates a servo motor to align the water nozzle and energizes a relay module that switches on a DC water pump.

This mechanism allows water to be sprayed in a targeted manner at the fire source. The block diagram encapsulates the entire signal flow from fire detection to suppression while ensuring real-time communication, precise actuation and remote control capability, making the robot both autonomous and user-controllable. This integrated design provides a reliable, efficient, and safe solution for small-scale firefighting tasks.

Flow Chart :-

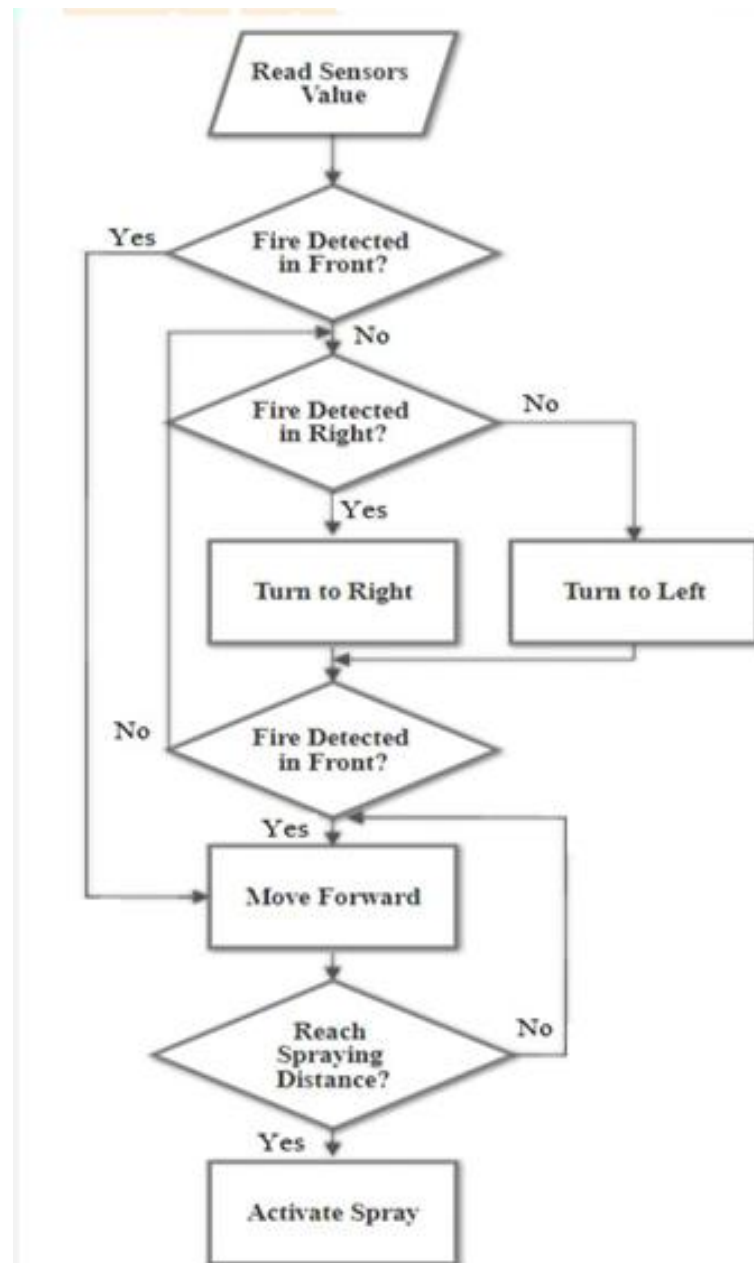
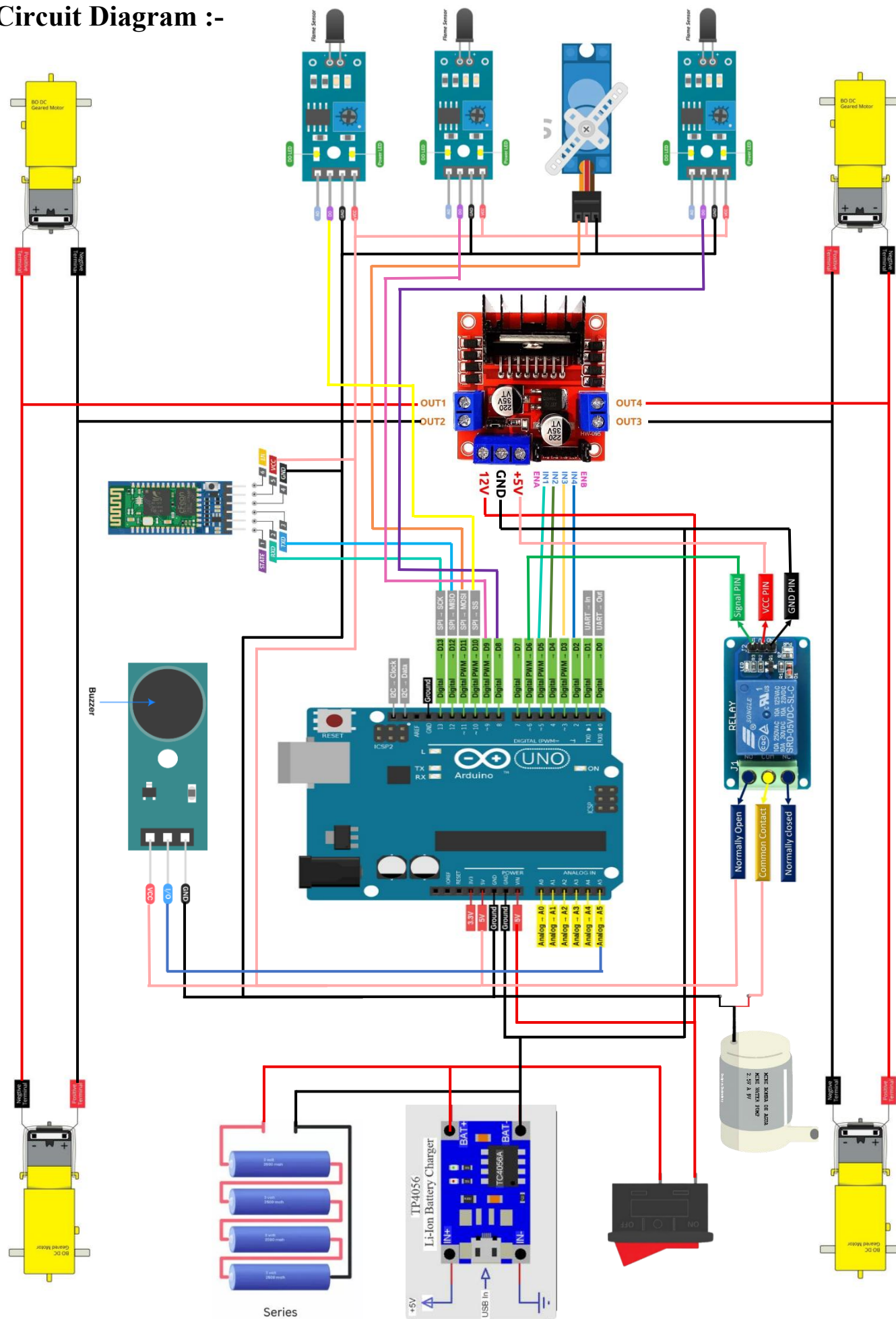


Fig 2 : Flow Chart

Circuit Diagram :-



Required Component & Tool with Quantity-

Sr. No.	Components	Specification	Quantity
1.	Arduino Uno	ATmega328P microcontroller	1
2.	Battery charging module	TP4056 o/p 12V,4A	1
3.	Bluetooth Module	HC-05 5V	1
4.	BO Moter	12V	4
5.	Buzzer	8850 5V	1
6.	Cable Tie	-	20
7.	Glue Gun	220V, 1A	1
8.	Glue Sticks	-	5
9.	Flame Sensor	Infrared 5V 60°	1
10.	LED	3V	4
11.	Lithium Battery	3.7V, 2000mAh	4
12.	Moter driver	L298N H-Bridge	1
13.	Multi-meter	Digital Multi-meter	1
14.	Nozzle	-	1
15.	PCB	-	1
16.	Ply wood	30*30cm	1m×1m
17.	Resistance	(1K)	4
18.	Scale	30cm	1
19.	Screwdriver	Tester	1
20.	Servo Moter	5V, 1mA 180°	1
21.	Single Channel Relay	5V	1
22.	Soldering Iron	60W, 220V	1
23.	Soldering Wire & Wick	0.25mm	1m
24.	Rocker Switch	12V On-Off	1
25.	Tweezer	Insulated	1
26.	Type C Charger	min- 20W	1
27.	Water Pipe	-	1m
28.	Water Pump	5V, 1A	1
29.	Water storage tank	1L	1
30.	Wheels	Rubber BO Wheel	4
31.	Wire Stepper	Insulated	1
32.	Wooden cutter	Aari Blead	1

Electronics Components

1. ARDUINO UNO

A) Introduction :-

The **Arduino UNO** is a Standard board of Arduino. Here mean one in Italian. It was named as UNO to label the first release of Arduino software. It was also the first USB board released by Arduino. It is considered as the powerful board used in various projects. Arduino. cc developed the Arduino UNO board.

Arduino UNO is based on an ATmega328P **microcontroller**. It is easy to use compared to other boards, such as the Arduino Mega board, etc. The board consists of digital and analog Input/Output pins (I/O), shield and other circuits. It is seen in figure 1.1

The **Arduino UNO** include 6 analog pin inputs 14 digital pins, a **USB connector**, a power jack, and an **ICSP (In-Circuit Serial, Programming)** header. It is programmed based on IDE, which stands for **Integrated development Environment**. It can run on both online and offline platform. Arduino makes a different type of development boards and each having different capabilities. Like **Arduino Mega**, **Arduino UNO**, **Arduino Nano**, **Lily Pad Arduino**, and much more because as advance our projects are, Arduino keeps on updating its development boards to keep teacher, students and other interest people updated as these boards are being using in scientific research as well as many other advanced fields of robotics and many other fields. These boards are commonly used as they are inexpensive, cross-platform (IDE software which is used to make a code for these development boards can run on Linux, SOX and Windows operating system while some boards are limited only to Window.

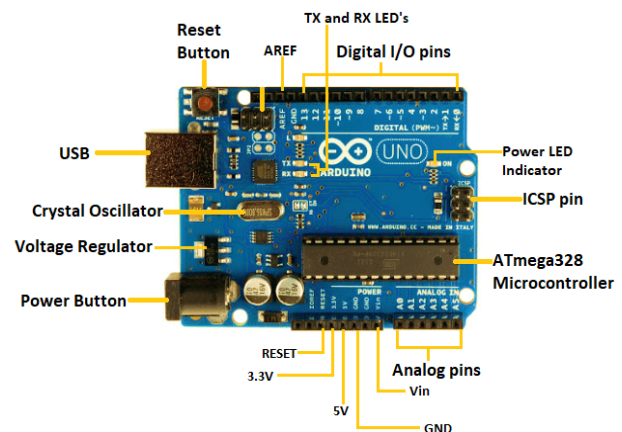


Fig. 1.1

B) Working :-

To work with an Arduino Uno, you follow a three-part process: setting up the hardware, writing code in the software, and uploading that code to the board.

1. **Hardware Setup**

The Arduino Uno is a microcontroller board that acts as the "brain" for electronic projects.

- **Connect to Power:** Use a USB A to B cable (the standard "printer cable") to connect the Uno to your computer. The USB port provides both power and a data connection for programming.
- **External Power (Optional):** For mobile projects, you can use a 9V battery or an AC adapter (7V–12V) via the DC power jack.
- **Pin Basics:**
 - a) **Digital Pins (0–13):** Used for on/off signals (e.g., turning on an LED or reading a button).
 - b) **Analog Pins (A0–A5):** Used to read variable signals from sensors like light or temperature.
 - c) **Power Pins:** Provide 5V or 3.3V to power external components, with GND (Ground) pins to complete the circuit.

2. Software Setup

You need the Arduino IDE (Integrated Development Environment) to write and upload code.

- Download: Get the latest version from the official Arduino Software page.
- Configure: Once open, go to the Tools menu:
 1. Board: Select Arduino Uno.
 2. Port: Select the serial port that appears when you plug in your board (often labeled "USB").

3. Programming (Writing a "Sketch")

Arduino programs are called sketches and are written in a simplified version of C++. Every sketch has two essential parts:

- void setup(): Code here runs once when the board starts (used to define pin modes).
- void loop(): Code here runs repeatedly in a loop until power is disconnected.

Your First Test: "Blink"

1. Go to File > Examples > 01.Basics > Blink.
2. Click the Verify button (checkmark icon) to check for errors.
3. Click the Upload button (right-arrow icon) to send the code to your board.
4. If successful, a small built-in LED labeled "L" on the board will start blinking.

Common Components for Beginners

- Breadboard: Used to prototype circuits without soldering.
- Resistors: Essential to limit current and protect components like LEDs from burning out.
- Jumper Wires: Used to connect the Arduino pins to components on the breadboard.

C) Arduino UNO Pin Diagram :-

The Arduino UNO board can be built with power pins, analog, ATmega328P ICSP header, reset button, **Power LED**, power supply.

The Arduino UNO board description is discussed below.

- **Power Supply:-** The Arduino UNO power supply can be done with the help of a USB cable or an External Power Supplies mainly include AC to DC adapter otherwise a battery. The adapter can be connected to the Arduino UNO by plugging into the Arduino board. Similarly, the battery lead can be connected to the Vin pin and the GND pin of the power connector. The suggested voltage range will be 7 Volt to volts.
- **Input & Output:-** The 14 digital pins on the Arduino UNO can be used as input & output with the help of the function like pin mode(), digital Write(),&Digital Read().

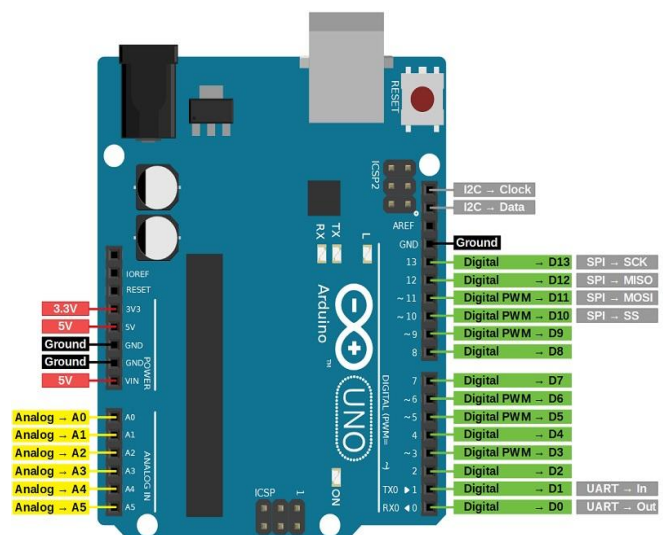


Fig. 1.2

- **Pin1(TX)&Pin0(RX) (Serial):-** This pin is used to transmit &receive TTL serial data, and these are connected to the ATmega8U2 USB to TTL serial chip equivalent pins.
- **Pin 2 & Pin 3 (External Interrupts):-** External pins can be connected to activate an interrupt over a low value, change in value.
- **Pins 3,5,6,9,10&11(PWM):-** This pin gives 8-bit PWM O/P by the function of analog Write.
- **Pin-12(MISO), Pin-13(SCK):-** These pins maintain SPI-communication, even though offered by the fundamental hardware, is not presently included within the Arduino language.
- **Pin-13(LED):-** The inbuilt LED can be connected to pin-13(digital pin). As the HIGH-value pin, the light emitting Diode is activated, whenever the pin is LOW.
- **Pin-4(SDA)&Pin-5(SCL):-** It supports TWI-communication with the help of Wire library.
- **AREF(Reference Voltage):-** The reference voltage is for the analog I/Ps with analog Reference.
- **Reset Pin:** This pin is used for reset (RST) the microcontroller.
- **Memory:-** The memory of this ATmega328 Arduino microcontroller includes flash memory-32kb for storing code, SRAM-2KB EEPROM-1KB.
- **ICSP Pin:-** The In-Circuit serial programming pin allows the user to program using the firmware of the Arduino board.
- **Power LED Indicator:-** The ON status of LED shows the power is activated. When the power is OFF, the LED will not light up.
- **Reset Button:-** It is used to add a reset button to the connection.
- **USB:-** It allows the board to connection to the computer. It is essential for the programming of the **Arduino UNO board**.
- **Voltage Regulator:-** The voltage regulator converts the input voltage to 5v.
- **Crystal Oscillator:-** The crystal Oscillator has a frequency of **16MHz**, which make the Arduino UNO a power board.
- **Vin:-** It is the input voltage.
- **Analog Pins:-** The pins numbered from A0 to A5 are analog pins. The functions of Analog pins. The function of analog pins is to read the analog sensor used in the connection. It can also act as **GPIO (General Purpose Input Output)** pins.
- **GND:-**Ground pins. The ground pin acts as a pin with zero voltage.

D) Applications of the Arduino UNO

- ♣ Security and defense.
- ♣ Embedded system.
- ♣ Digital Electronics and Robotics.
- ♣ Weighing machines.
- ♣ Parking Lot Counter.
- ♣ Medical instrument.
- ♣ Traffic Light Counter Down Timer.
- ♣ Home Automation.
- ♣ Industrial Automation.
- ♣ Emergency Light for Railways.

2. L298N Motor Driver :-

This dual bidirectional motor driver, is based on the very popular L298 Dual H-Bridge Motor Driver Integrated Circuit. The circuit will allow you to easily and independently control two motors of up to 2A each in both directions. It is ideal for robotic applications and well suited for connection to a microcontroller requiring just a couple of control lines per motor. It can also be interfaced with simple manual switches, TTL logic gates, relays, etc. This board equipped with power LED indicators, on-board +5V regulator and protection diodes. We can see in figure 2.1

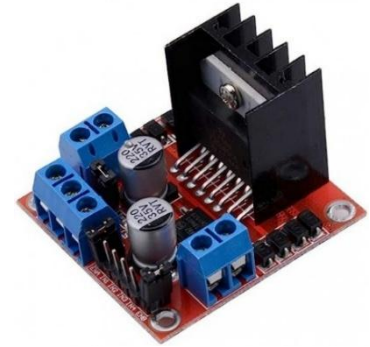


Fig. 2.1

A) Technical Specifications:

- **Operating Voltage (VCC):** 3.2V to 40V (motor supply voltage)
- **Logic Voltage (VSS):** 5V to 7V (5V typically used from Arduino)
- **IC Used:** L293D Dual H-Bridge Motor Driver IC
- **Peak Current :** 2 Amp
- **Operating current range :-** 0~36 mA
- **Motor Channels:** 2 (can control 2 DC motors or 1 stepper motor)
- **Control Pins:** IN1, IN2 (Motor A), IN3, IN4 (Motor B), ENA/ENB for enabling or PWM speed control
- **Output Pins:** OUT1–OUT4 (connect to motors)
- **Speed Control (PWM):** Yes, via ENA and ENB pins (connect to Arduino PWM pins)
- **Built-in Protection:** Internal diodes for back EMF, thermal shutdown protection
- **Current per Channel:** 600 mA continuous, 1 A peak per motor channel

B) Schematic Diagram:

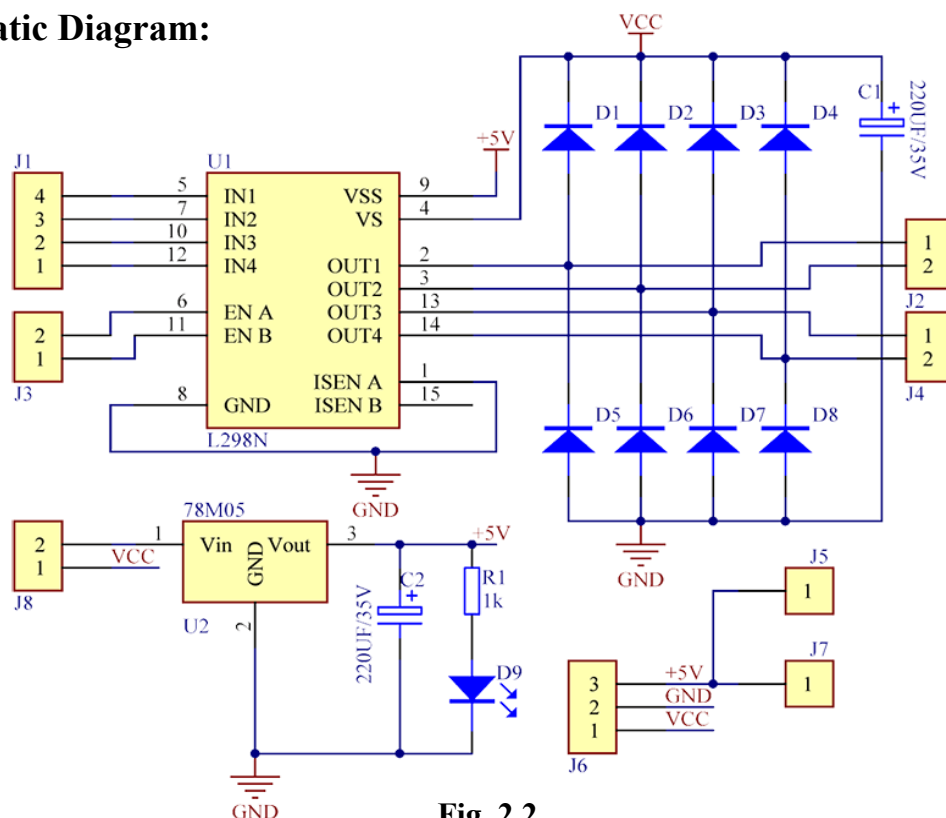


Fig. 2.2

C) Pin Diagram & their Specification

IN1 & IN2 :- Motor A input pins. Used to control the spinning direction of Motor A

IN3 & IN4 :- Motor B input pins. Used to control the spinning direction of Motor B

ENA :- Enables PWM signal for Motor A

ENB :- Enables PWM signal for Motor B

OUT1 & OUT2 :- Output pins of Motor A

OUT3 & OUT4 :- Output pins of Motor B

12V :- 12V input from DC power Source

5V :- Supplies power for the switching logic circuitry inside L298N IC

GND :- Ground Pin

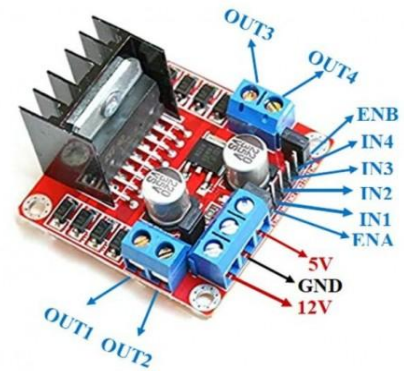


Fig. 2.3

D) Controlling 2-DC Motor with +5V Arduino onboard Power Supply:

Below is the circuit connection use the on-board +5V power supply from Arduino board, and should be done without the 5V Enable Jumper on (Active 5V). This connection can drive two 5V DC motors simultaneously.

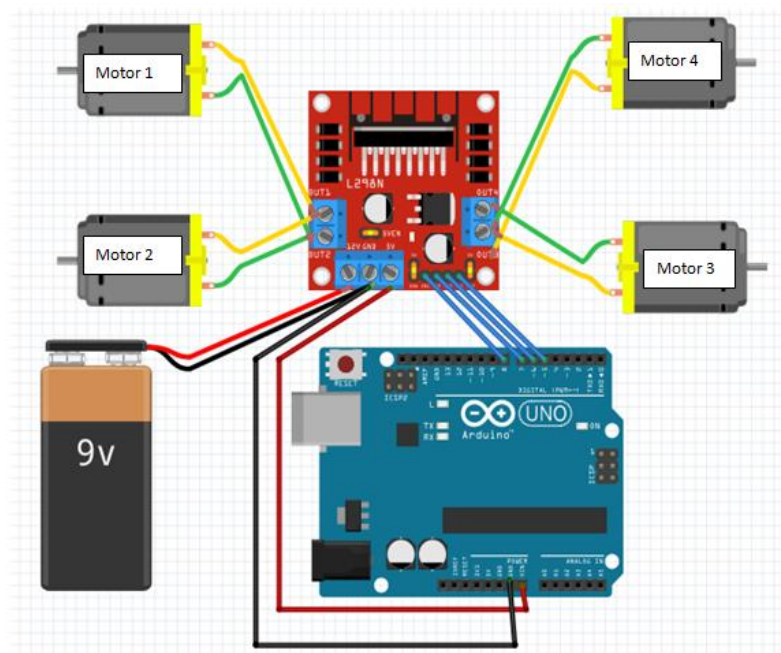


Fig. 2.4

www.roboticsinsightto.com

Applications

- Drive DC motors.
- Drive stepping motors
- In Robotics

3. Light emitting diode [LED]

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it. When current passes through an LED, the electrons recombine with holes emitting light in the process. LEDs allow the current to flow in the forward direction and blocks the current in the reverse direction.

Light-emitting diodes are heavily doped p-n junctions. Based on the semiconductor material used and the amount of doping, an LED will emit coloured light at a particular spectral wavelength when forward biased. As shown in the figure, an LED is encapsulated with a transparent cover so that emitted light can come out.

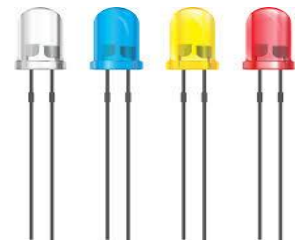


Fig. 3.1

- **LED Symbol:-** The LED symbol is the standard symbol for a diode, with addition of two small arrows denoting the emission of light. We can see in figure 3.2.



the

Fig. 3.2

- **Simple LED Circuit:** The figure 3.3 below shown a simple led circuit. The circuit consists of an LED, a voltage supply and a resistor to regulate the current and voltage.

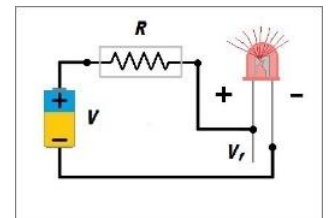


Fig. 3.3

4. BO - MOTER :-

BO (Battery Operated) motors are lightweight, 3-12V DC geared motors designed for robotics, typically offering 60-600 RPM speeds, roughly 1Kg-cm torque, and high current efficiency. These popular yellow-cased, plastic-geared motors feature single or dual-shaft designs, making them ideal for small, cost-effective, battery-powered hobby projects.

Typical BO Motor Specifications:

- **Operating Voltage:** 3V – 12V DC (6V–9V recommended for most)
- **No Load Current:** ~40-180 mA
- **Stall Current:** Up to ~600 mA
- **Speed (RPM):** Available in 60, 100, 150, 300, and 500+ variants
- **Output Torque:**
- **Shaft Type:** Double D-type, Dual or Single shaft
- **Shaft Diameter:** 5.5 mm
- **Weight:** ~30-50 grams



Fig. 4.1

Key Features:

- **High Gear Ratio:** Provides high torque in a small form factor, perfect for robotic wheels.
- **Compatibility:** Easily fits standard plastic BO wheels and connects with motor drivers like L298N.
- **Design:** Compact, lightweight, with integrated mounting holes for easy mounting on a chassis.
- **Variants:** Available in both straight and L-shape (right angle) output configurations.

5. FLAME SENSER:-

A flame-sensor is one **kind of detector** which is mainly designed for detecting as well as responding to the occurrence of a fire or flame. The flame detection response can depend on its fitting. It includes an **alarm system**, a natural gas line, propane & a fire suppression system. This sensor is used in **industrial boilers**. The main function of this is to give authentication whether the boiler is properly working or not. The response of these sensors is faster as well as more accurate compare with a heat/smoke detector because of its mechanism while detecting the flame.

A sensor which is most sensitive to a normal light is known as a flame sensor. That's why this **sensor module** is used in flame alarms. This sensor detects flame otherwise wavelength within the range of 760 nm – 1100 nm from the light source. This sensor can be easily damaged to high temperature. So, this sensor can be placed at a certain distance from the flame. The flame detection can be done from a 100cm distance and the detection angle will be 600. The output of this sensor is an analog signal or digital signal. These sensors are used in firefighting robots like as a flame alarm. Its also see in figure 5.1

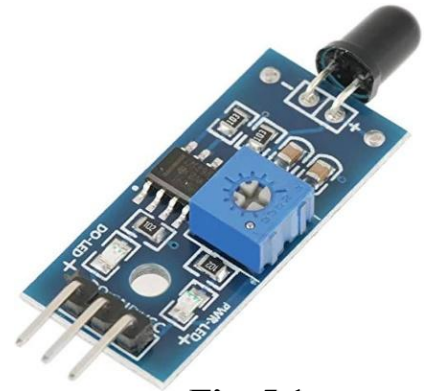


Fig. 5.1

Working:

This sensor/detector can be built with an **electronic circuit** using a receiver like electromagnetic radiation. This sensor uses the infrared flame flash method, which allows the sensor to work through a coating of oil, dust, water vapor, otherwise ice.

The pin configuration of this sensor is shown below. It includes four pins which include the following.

When this module works with a microcontroller unit then the pins are shown in below fig. 5.2

- **Pin1 (VCC pin):** Voltage supply ranges from 3.3V to 5.3V
- **Pin2 (GND):** This is a ground pin
- **Pin3 (AOUT):** This is an analog output pin (MCU.IO)
- **Pin4 (DOUT):** This is a digital output pin

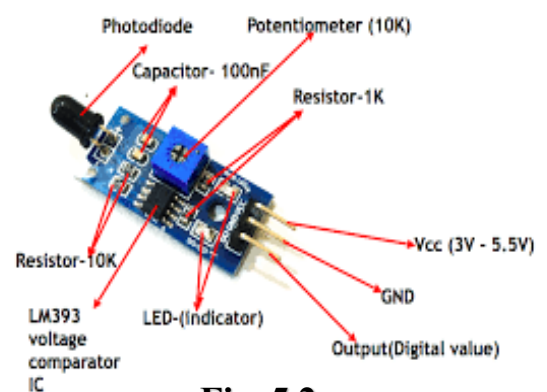


Fig. 5.2

Different Types :- Flame-sensors are classified into four types

- IR single frequency
- IR multi-spectrum
- UV flame detectors
- UV/ IR flame detectors

Features & Specifications:- The features of this sensor include the following.

- Photosensitivity is high
- Response time is fast
- Simple to use
- Sensitivity is adjustable
- Detection angle is 600,
- It is responsive to the flame range.
- Accuracy can be adjustable
- Operating voltage of this sensor is 3.3V to 5V
- Analog voltage o/ps and digital switch o/ps
- The PCB size is 3cm X 1.6cm
- Power indicator & digital switch o/p indicator
- If the flame intensity is lighter within 0.8m then the flame test can be activated, if the flame intensity is high, then the detection of distance will be improved.

Applications:- These sensors are used in several dangerous situations which include the following.

- Hydrogen stations
- Industrial heating
- Fire detection
- Fire alarm
- Firefighter robot
- Drying systems
- Industrial gas turbines
- Domestic heating systems
- Gas-powered cooking devices

6. Bluetooth HC-05 :-

The HC-05 is a popular, high-performance Bluetooth SPP (Serial Port Protocol) module designed for transparent wireless serial connections. It operates on Bluetooth v2.0+EDR (2.4GHz) and supports both Master and Slave modes, allowing it to initiate or receive connections. It typically uses 3.6V-6V power, communicates via UART (9600-38400 baud), and has a ~10-meter range.

Wireless communication is swiftly replacing the wired connection when it comes to electronics and communication. Designed to replace cable connections HC-05 uses serial communication to communicate with the electronics. Usually, it is used to connect small devices like mobile phones using a short-range wireless connection to exchange files. It uses the 2.45GHz frequency band. The transfer rate of the data can vary up to 1Mbps and is in range of 10 meters.

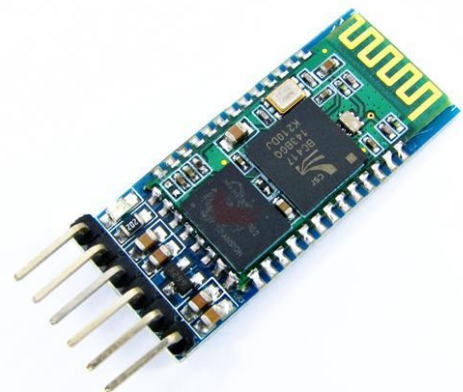


Fig. 6.1

The HC-05 module can be operated within 4-6V of power supply. It supports baud rate of 9600, 19200, 38400, 57600, etc. Most importantly it can be operated in Master-Slave mode which means it will neither send or receive data from external sources.

Bluetooth serial modules allow all serial enabled devices to communicate with each other using Bluetooth.

It has 6 pins,

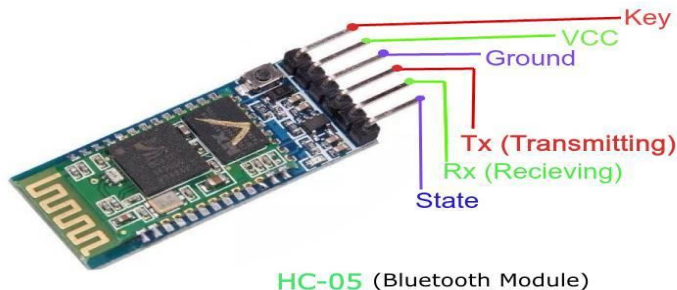


Fig. 6.2(a)



Fig. 6.2(b)

1. Key/EN: It is used to bring Bluetooth module in AT commands mode. If Key/EN pin is set to high, then this module will work in command mode. Otherwise by default it is in data mode. The default baud rate of HC-05 in command mode is 38400bps and 9600 in data mode.

HC-05 module has two modes,

- Data mode: Exchange of data between devices.
- Command mode: It uses AT commands which are used to change setting of HC-05. To send these commands to module serial (USART) port is used.

2. VCC: Connect 5 V or 3.3 V to this Pin.

3. GND: Ground Pin of module.

4. TXD: Transmit Serial data (wirelessly received data by Bluetooth module transmitted out serially on TXD pin)

5. RXD: Receive data serially (received data will be transmitted wirelessly by Bluetooth module).

6. State: It tells whether module is connected or not.

Working:-

Modes of Operation

The HC-05 Bluetooth Module can be used in two modes of operation: Command Mode and Data Mode.

Command Mode

In Command Mode, you can communicate with the Bluetooth module through AT Commands for configuring various settings and parameters of the Module like get the firmware information, changing Baud Rate, changing module name, it can be used to set it as master or slave.

A point about HC-05 Module is that it can be configured as Master or Slave in a communication pair. In order to select either of the modes, you need to activate the Command Mode and sent appropriate AT Commands.

Data Mode

Coming to the Data Mode, in this mode, the module is used for communicating with other Bluetooth device i.e. data transfer happens in this mode.

- HC-05 has red LED which indicates connection status, whether the Bluetooth is connected or not. Before connecting to HC-05 module this red LED blinks continuously in a periodic manner. When it gets connected to any other Bluetooth device, its blinking slows down to two seconds.
- This module works on 3.3V. We can connect 5V supply voltage as well since the module has on board 5 to 3.3 V regulator.
- As HC-05 Bluetooth module has 3.3V level for RX/TX and microcontroller can detect 3.3 V level, so, no need to shift transmit level of HC-05 module. But we need to shift the transmit voltage level from microcontroller to RX of HC-05 module.
- The data transfer rate of HC-05 module can vary up to 1Mbps is in the range of 10 meters.

Pair HC-05 and smartphone:

1. Search for new Bluetooth device from your phone. You will find Bluetooth device with “HC-05” name.
2. Click on connect/pair device option; default pin for HC-05 is 1234 or 0000.

After pairing two Bluetooth devices, open terminal software (e.g. Teraterm, Realterm etc.) in PC, and select the port where we have connected USB to serial module. Also select default baud rate of 9600 bps.

In smart phone, open Bluetooth terminal application and connect to paired device HC-05.

It is simple to communicate, we just have to type in the Bluetooth terminal application of smartphone. Characters will get sent wirelessly to Bluetooth module HC-05. HC-05 will automatically transmit it serially to the PC, which will appear on terminal. Same way we can send data from PC to smartphone.

Alternate options for HC-05 Bluetooth Module

1. HC-06 Bluetooth module: This is a similar module to the HC-05, but it is limited to a slave role only. It has a smaller form factor and is generally cheaper than the HC-05. However, it does not support some of the advanced features of the HC-05, such as the ability to enter AT mode to configure the module.
2. HM-10 Bluetooth module: This is a more advanced Bluetooth module that supports Bluetooth 4.0 (BLE) and can act as both a master and slave device. It also supports a wider range of AT commands for configuring the module, and has a longer range than the HC-05. However, it is generally more expensive than the HC-05.
3. RN-42 Bluetooth module: This is another Bluetooth module that supports both the SPP and HID profiles, similar to the HC-05. It has a longer range than the HC-05 and supports faster data rates. However, it is also more expensive and may require additional configuration to work properly.
4. ESP32 Bluetooth module: This is a powerful Wi-Fi and Bluetooth module that includes a dual-core processor and support for both Bluetooth Classic and BLE. It is more expensive than the HC-05, but offers more advanced features and capabilities.
5. nRF24L01+ Wireless module: This is a wireless module that operates at 2.4GHz and uses a different protocol than Bluetooth. It is generally cheaper than Bluetooth modules and can be used for applications where a shorter range and lower data rate are acceptable.

Key Technical Specifications:

- **Bluetooth Protocol:** Bluetooth v2.0+EDR
- **Operating Frequency:** 2.4GHz ISM band
- **Modulation:** GFSK (Gaussian Frequency Shift Keying)
- **Emission Power:** 4dbm, Class 2
- **Sensitivity:** -84dbm at 0.1% BER
- **Operating Voltage:** 3.6V to 6V DC (3.3V recommended logic)
- **Working Current:** 30mA (pairing), 8mA (communication)
- **Range:** Typical 10 meters
- **UART Baud Rate:** 9600, 19200, 38400, etc. (Default: 9600)
- **Operating Temperature:** -20 to +75 °C
- **Dimensions:** 26.9mm x 13mm x 2.2mm

Key Features:

- **Master/Slave Capability:** Configurable to act as Master, Slave, or Loopback.
- **AT Commands:** Features built-in AT commands for configuration, such as changing baud rate or device name.
- **6-Pin Configuration:** Typically includes VCC, GND, TXD, RXD, State, and Button (Key) pins.
- **Secure Connection:** Supports authentication and encryption.

Application:

- Computer and peripheral devices
- GPS receiver
- Industrial control
- MCU projects

Pin Diagram of Relay and their specification :-

The pin configuration of the 5V relay module is shown below. This module includes 6-pins where each pin and its functionality are discussed below & see in fig 7.4.

Normally Open (NO): This pin is normally open unless we provide a signal to the relay modules signal pin. So, the common contact pin smashes its link through the NC pin to make a connection through the NO pin.

Common Contact: This pin is used to connect through the load that we desire to switch by using the module.

Normally Closed (NC): This NC pin is connected through the COM pin to form a closed circuit. However, this NC connection will break once the relay is switched through providing an active high/low signal toward the signal pin from a microcontroller.

Signal Pin: The signal pin is mainly used for controlling the relay. This pin works in two cases like active low otherwise active high. So, in active low case, the relay activates once we provide an active low signal toward the signal pin, whereas, in an active high case, the relay will trigger once we provide a high signal toward the signal pin.

However, these modules generally work on an active high signal which will strengthen the relay coil to make contact with the common terminal with the normally open terminal.

5V VCC: This pin needs 5V DC to work. So 5V DC power supply is provided to this pin.

Ground: This pin connects the GND terminal of the power supply.



Fig. 7.4

Specifications:-

- Supply voltage – 3.75V to 6V
- Quiescent current: 2mA
- Current when the relay is active: ~70mA
- Relay maximum contact voltage – 250VAC or 30VDC
- Relay maximum current – 10A

Single-Channel Relay Module Basic Trouble Shooting :- If the relay does not switch on, i.e. no audible clicking sound is heard:

- The contacts might be stuck - Check by physically shaking the relay, if a light clicking sound is not heard, then tap the relay hard, in most cases, this should 'unstick' both the contacts.
- If the contacts do click when the relay is shaken, then the transistor or the flyback diode might be damaged and must be replaced.

Applications :-

- Mains switching
- High current switching
- Isolated power delivery
- Home automation

8. Servo motor :-

The SG90 is a 9-gram **micro servo motor** made by Tower Pro. Unlike DC motors (which spin continuously) or stepper motors (which move in discrete steps), a servo motor moves to a **specific angle** and holds that position controlled by a PWM signal from your Arduino. The “9g” refers to its weight: just 9 grams, making it extremely lightweight. It’s shown in figure 8.1.

Inside the SG90 is a DC motor, a gearbox (made of plastic in the SG90), a position feedback potentiometer, and a control circuit all packed into a housing smaller than a matchbox. This combination allows precise position control with minimal code.



Fig. 8.1

Technical Specifications :-

Weight	9 grams
Dimensions	22.2mm × 11.8mm × 31mm
Operating voltage	4.8V – 5V (up to 6V max)
Stall torque at 4.8V	1.8 kg·cm (17.7 N·cm)
Stall torque at 6V	2.2 kg·cm (21.6 N·cm)
No-load speed at 4.8V	0.1 sec/60° (100 RPM)
Rotation range	180° (0° to 180°)
Gear type	Plastic gears
Control signal	PWM (50Hz, 1ms–2ms pulse)
Dead bandwidth	10μs
Operating temperature	0°C to 55°C
Connector	JR/Futaba style 3-pin

SG90 Pinout (Wire Colours) :-

Wire Colour	Function	Connect To
Brown (or Black)	Ground (GND)	Arduino GND
Red	Power (VCC)	Arduino 5V (or external 5V)
Orange (or Yellow/White)	Signal (PWM)	Arduino digital pin (e.g., Pin 9)

Servo Motor Works :-

Understanding the control mechanism of a servo makes programming it much more intuitive.

PWM Control Signal

Servo motors are controlled by a Pulse Width Modulation (PWM) signal at 50Hz (one pulse every 20ms).

The *width* of the pulse determines the servo's position:

- 1ms (1000µs) pulse: Servo moves to 0°
- 1.5ms (1500µs) pulse: Servo moves to 90° (Centre position)
- 2ms (2000µs) pulse: Servo moves to 180°

The servo has a built-in feedback potentiometer connected to the output shaft. The internal circuit compares the commanded position (from the PWM pulse width) to the actual position (from the potentiometer) and drives the motor until they match. This is called a closed-loop position control system.

Why Not Just Use Analog Write()?

Arduino's `Analog Write()` generates PWM at 490Hz or 980Hz (not 50Hz) and at 8-bit resolution. Servo motors need 50Hz PWM with specific pulse widths. Arduino's Servo library handles all this automatically, using hardware timer interrupts to generate the correct signal. This is why you use `servo.write(angle)` instead of `Analog Write()`

Common Applications

- **Robotics:** Small robot arms, grippers, and walking bipeds.
- **RC Vehicles:** Steering in RC cars, and controlling ailerons or elevators in RC planes.
- **DIY Projects:** Camera gimbals (pan-tilt mounts), automated door locks, and pet feeders.

Usage Tips

- **Separate Power:** For projects using more than two servos, use an **external 5V power supply** rather than drawing directly from an Arduino pin to avoid stability issues.
- **Avoid Manual Rotation:** Do not turn the motor shaft by hand, as this can easily strip the internal plastic gears.

9. Lithium Battery :-

Lithium-ion (Li-ion) batteries typically feature a nominal voltage of 3.6V–3.7V per cell, with capacities ranging from hundreds of mAh to over 100Ah depending on size (e.g., 18650, 21700, prismatic). Key specs include 100–250 Wh/kg specific energy, 500–2000+ cycle life, and operating temperatures from -20°C to +60°C. Its shown in figure 9.1.



Fig. 9.1

Common Technical Specifications :-

- **Nominal Voltage:** 3.6V or 3.7V per cell, operating down to 3.0V and charging up to 4.2V.
- **Capacity:** Measured in mAh (small cells) or Ah (large packs). Example: 18650 cells typically offer 2500–3500mAh.
- **Cycle Life:** 500–1000+ cycles, with newer Lithium Iron Phosphate (LiFePO₄) often exceeding 2,000+ cycles.
- **Energy Density:** Varies by chemistry, ranging from ~100 Wh/kg to over 200 Wh/kg.
- **Charge/Discharge Rate (C-rate):** Standard charge is often 0.5C–1C, with discharge capabilities usually up to 1C–3C continuously (up to 30C pulse for specialized cells).
- **Safety Features:** Requires a Battery Management System (BMS) to monitor voltage, current, and temperature (thermal runaway risks occur around 150°C–270°C).

Common Cell Formats :-

- **Cylindrical:** 18650 (18mm x 65mm), 21700 (21mm x 70mm), 32650 (32mm x 65mm).
- **Prismatic:** Rectangular, often used in electric vehicles and larger storage units.
- **Pouch:** Flexible casing commonly found in smartphones and laptops.

Temperature Ranges :-

- **Charging:** 0°C to +45°C (lower temperatures can damage cells).
- **Discharging:** -20°C to +60°C.

Specifications for Common Chemistries :-

- **Li-cobalt (LCO):** 150–190Wh/kg, 500 cycles.
- **Li-manganese (LMO):** 100–135Wh/kg, 500–1,000 cycles.
- **Li-phosphate (LFP):** 90–120Wh/kg, 1,000–2,000 cycles.
- **NMC (Nickel Manganese Cobalt):** 140Wh/kg, 1,000–2,000 cycles.

Safety and Environmental Specs :-

- **Operating Temperature:** Typically -20°C to 60°C for discharge; charging is usually restricted to 0°C to 45°C to prevent damage.
- **Self-Discharge Rate:** Approximately 1.5% to 3% per month, significantly lower than nickel-based alternatives.
- **Internal Resistance:** Usually measured in milliohms (mΩ); lower resistance indicates better health and higher power delivery.

10. Resistance :-

The primary specification for a resistor is its resistance value, measured in ohms (Ω), which indicates how much it opposes the flow of electric current. However, a complete resistance specification includes several other critical parameters that determine its performance and safety in a circuit.

Key Specifications :-

- **Resistance Value:** The nominal ohmic value, often denoted by a 3-4 digit code or color bands (e.g., E24 or E96 series).
- **Tolerance:** The allowable percentage deviation from the nominal value, typically ranging from $\pm 0.1\%$ (high precision) to $\pm 5\%$ or $\pm 10\%$ (general purpose).
- **Power Rating (Wattage):** Maximum power (heat) a resistor can dissipate without damage (e.g., $1/8W$, $1/4W$, $1W$, $5W$, $10W$). It depends on ambient temperature and physical size.
- **Temperature Coefficient of Resistance (TCR):** Expressed in $\text{ppm}/^\circ\text{C}$, this measures how much the resistance value changes with temperature variations.
- **Operating Voltage:** The maximum continuous voltage that can be applied to the resistor.
- **Mounting/Package Type:** Defines how the resistor is attached to the circuit, such as Surface Mount Device (SMD) for low power or through-hole (axial) for higher power.

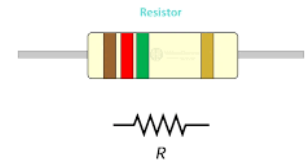
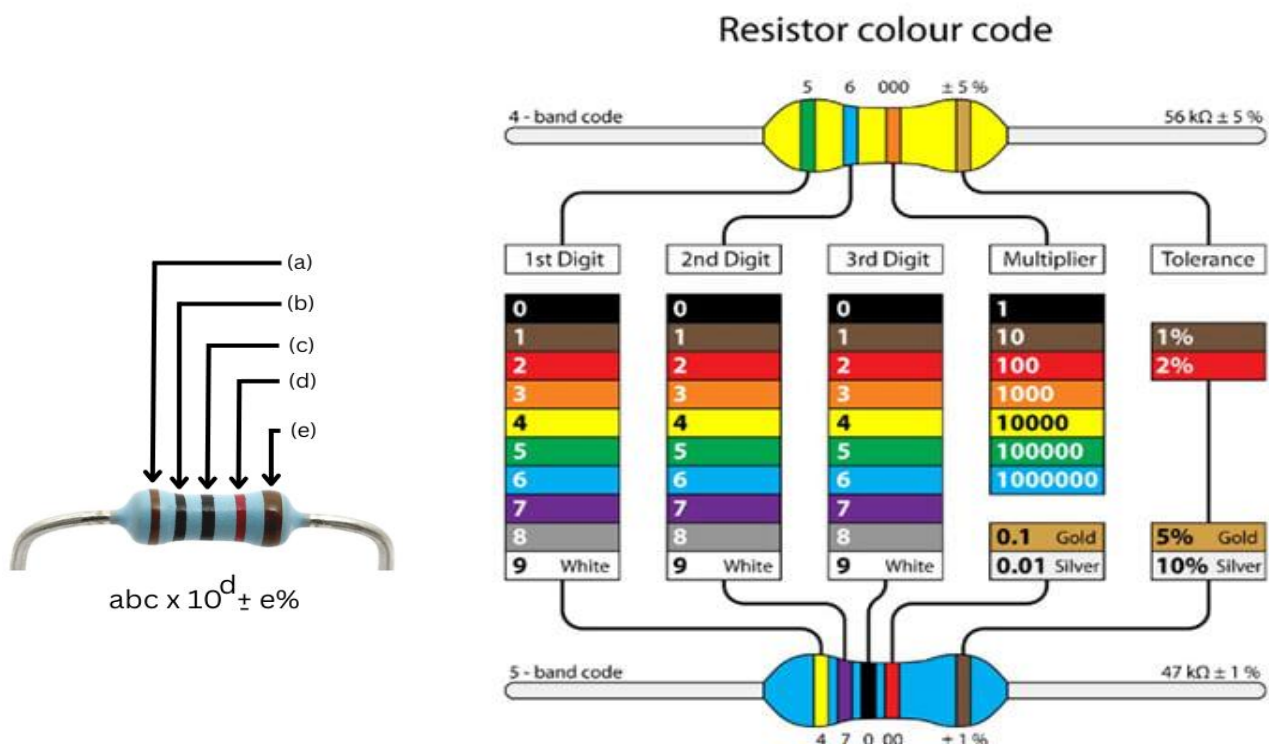


Fig. 10.1

Operational Factors :-

- **Operating Temperature Range:** Safe, reliable temperature limits.
- **Power Derating:** Reduction of power handling at higher temperatures.
- **Frequency Response:** Performance limits due to parasitic effects.
- **Standards:** Commonly determined by IEC 60062 and IEC 60063.

Resistance Colour Coding :-



11. Battery Charging Module :-

The TP4056 is a popular, low-cost linear charging module designed for single-cell 3.7V lithium-ion (Li-ion) or lithium-polymer (LiPo) batteries, such as 18650 cells. It typically supports a maximum charging current of 1A and operates via a USB interface (Micro, Mini, or Type-C) or direct DC input.

Key Specifications :-

- **Input Voltage:** 4.5V to 5.5V.
- **Full Charge Voltage:** 4.2V ($\pm 1.5\%$ accuracy).
- **Charging Current:** 1A by default, but programmable from 100mA to 1000mA by changing the RPROG (R3) resistor.
- **LED Indicators:** Typically, Red for charging and Green/Blue for fully charged.
- **Protection (for modules with DW01 IC):** Includes overcharge, over-discharge (cutoff at $\sim 2.4V$ – $2.5V$), and overcurrent ($\sim 3A$) protection.

Pinout & Connections :- Its shown in figure 11.1.

- **IN+ / IN-:** Alternative 5V DC input pads if not using USB.
- **B+ / B-:** Terminals for the battery's positive and negative leads.
- **OUT+ / OUT-:** Terminals for the external load

Common Issues :-

- **USB-C Compatibility:** Some cheap Type-C modules lack necessary resistors to work with C-to-C cables; they may only work with A-to-C cables.
- **Heat:** The chip can reach $\sim 60^{\circ}C$ during 1A charging, which is normal for a linear charger.

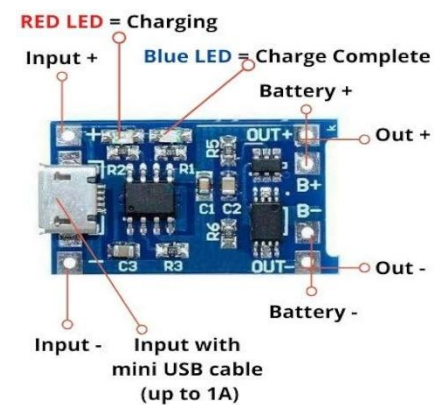


Fig. 11.1

12. PCB :-

A Printed Circuit Board (PCB) is a foundational electronic component that mechanically supports and electrically connects components—such as resistors, capacitors, and ICs—using conductive copper pathways etched onto non-conductive substrates like FR4 or polyimide. PCBs enable automated assembly and are essential to modern electronics, ranging from simple rigid single-layer boards to complex multilayer and flexible circuits. Its shown in figure 12.1.

Key Aspects of PCB Technology :-

- **Structure:** Comprises conductive copper layers laminated with insulating layers (e.g., fiberglass).
- **Key Components:** Active components (generate energy, e.g., ICs) and passive components (store energy, e.g., resistors) are soldered onto the board.
- **Design Process:** Created using EDA software (e.g., Ki-Cad, Autodesk Eagle, Easy-EDA) to generate Gerber files, which define the copper tracks, solder mask, and silkscreen.
- **Manufacturing:** Fabricated through processes like milling or chemical etching. Popular fabrication services include JLCPCB and OSH Park.



Fig. 12.1

- **Types:** Single-layer, double-layer, multi-layer, rigid, flexible, and rigid-flex, used in everything from computers to consumer appliances.

Core Anatomy :-

- **Substrate (Base Layer):** Typically made of FR-4 (fiberglass epoxy laminate), which provides structural integrity and insulation.
- **Copper Layer:** Thin foil laminated to the substrate, etched into "traces" that carry signals between components.
- **Solder Mask:** The protective (usually green) coating that insulates traces and prevents accidental short circuits.
- **Silkscreen:** A layer of printed labels and symbols used to identify component locations and orientations.

Common PCB Types :-

- **Single-Sided:** Components are mounted on one side; used in simple devices like calculators.
- **Double-Sided:** Features copper on both sides, allowing for higher circuit density in products like industrial controls.
- **Multi-Layer:** Contains three or more conductive layers sandwiched together; essential for complex hardware like smartphones and motherboards.
- **Flexible (Flex):** Built on flexible plastics (like polyimide) to fit into tight or moving spaces.
- **Aluminum-Backed:** Uses an aluminum base for superior heat dissipation, commonly found in LED lighting.

Popular Design & Fabrication Tools :-

- **Design (EDA):** Ki Cad, Autodesk Eagle, Easy EDA, Dip Trace.
- **Fabrication/Assembly:** JLCPCB, OSH Park, Lion Circuits (India).

13.Wheels :-

This is 65mm Robot Wheel for BO Motors (Yellow). The wheel is made up of high-quality rubber which gives maximum traction while operating. The wheel is strong and sturdy as it features a nylon reinforced plastic rim. It's shown in figure 13.1.

Specifications:-

- Body Material: Plastic
- Grip Material: Rubber
- Shaft Type: D-type
- Loading Capacity: 2Kg
- Wheel Diameter: 70mm
- Wheel Width: 10mm
- Color: Red
- Weight: 15gm



Fig. 13.1

Mechanical Component

14.WATER PUMP :-

Mini DC 3-6V Micro Submersible Pump Mini water pump For Fountain Garden Mini water circulation System DIY project. This is a low cost, small size Submersible Pump Motor which can be operated from a 3 ~ 6V power supply. It can take up to 120 liters per hour with very low current consumption of 220mA. Just connect tube pipe to the motor outlet, submerge it in water and power it. Make sure that the water level is always higher than the motor. Dry run may damage the motor due to heating and it will also produce noise. It is shown below figure 14.1.

- Used to pump water and abnormal condition arises in the system .
- When excess temperature ,smoke and flame detected ,the water pump start pumping water .
- Water pump is connected to the relay , the relay become energizes according to the instruction from the MCU .

Mini Water Pump Specifications:-

- Operating Voltage : 3 ~ 6V
- Operating Current : 130 ~ 220mA
- Flow Rate : 80 ~ 120 L/H
- Maximum Lift : 40 ~ 110 mm
- Continuous Working Life : 500 hours
- Driving Mode : DC, Magnetic Driving
- Material : Engineering Plastic
- Outlet Outside Diameter : 7.5 mm
- Outlet Inside Diameter : 5 mm



Fig. 14.1

15.BUZZER :-

A buzzer is an electro-acoustic transducer that converts DC electrical signals into sound (beeps or buzzing). It acts as an audio indicator in devices like alarm systems, timers, and electronics, often utilizing a piezoelectric element or electromagnet to vibrate a diaphragm. They operate between 3–24V DC.

An Active Buzzer Alarm Module for Arduino is an audio signaling device, which may be mechanical, electromechanical, or piezoelectric. Just like what you are viewing now, it is 3.3V-5V DC Electronic Part Active Buzzer Module. Using top quality material, it is durable in use. Its shown in figure 15.1.

An active buzzer rings out as long as it is electrified. Compared with a passive buzzer, it is a bit expensive but easier to control. Typical uses of buzzers include alarm devices, timers, and confirmation of user input such as a mouse click or keystroke.

Specifications of Active Buzzer Module:-

- Product Name: 3.3 to 5V Active Buzzer Alarm Module Sensor
- Transistor drive module uses 8550
- With fixed bolt hole- easy installation- 2.6mm aperture.
- Operating voltage 3.3V-5V
- PCB Dimensions: 34.28 mm (L) * 13.29 mm (W) * 11.5 mm (H)



Fig. 15.1

Features:-

Passive buzzer features are:-

- Passive internal shocks without source- so if you can't make it with a DC signal tweet. Must be a square wave 2K ~ 5K to drive it
- Sound frequency control- you can make a "more than a meter hair Suola" effect.
- In some special cases- you can reuse a control and LED mouth

Active buzzer features are:-

- An active buzzer with a concussion internal source- so long as it will be called an energized
- Program easy to control- SCM can let a high-low sound- while passive buzzer did not.

Module interface specification (3-wire) :-

- + External 3.3V-5V voltage (can be directly connected with the 5v and 3.3v MCU)
- - External GND out external microcontroller
- IO port for control

16. ROCKER SWITCH :-

A rocker switch is a common electrical switch that pivots (or "rocks") on a hinge to turn circuits on and off, with one side rising while the other is depressed. Often called a seesaw switch, it is commonly found on power strips, appliances, and automotive dashboards to control current flow. It's shown in figure 16.1.



Fig. 16.1

Key Characteristics and Types:-

- 1. Mechanism:** Switches operate via movable contacts, either manually (toggle, slide, push-button) or automatically (limit switch).
- 2. Poles and Throws:** Pole refers to the number of separate circuits controlled, and throw refers to the number of paths (positions).
 - **SPST (Single-Pole Single-Throw):** Basic on-off switch, like a light switch.
 - **SPDT (Single-Pole Double-Throw):** Changes current flow between two paths.

17.Nozzle

18.Plywood

19.Water Pipe

20.Water storage tank

21.Type C-Charger

Tools :-

- | | |
|-------------------------|---------------------|
| ❖ Wooden cutter | ❖ Tweezer |
| ❖ Wire Stepper | ❖ Multi-meter |
| ❖ Soldering Iron | ❖ Cable Tie |
| ❖ Soldering Wire & wick | ❖ Glue Gun & Sticks |
| ❖ Screw driver | |

WORKING OF FIRE FIGHTER ROBOT

A firefighter robot works by using onboard sensors typically IR flame sensors to detect fire, controlled by a microcontroller like Arduino. It navigates towards the flame using motor drivers, stops when close, and activates a water pump via a relay to extinguish the fire, often using a servo motor for precise spraying.

This video demonstrates the working of a firefighting robot with an Arduino:

Key Components and Working Principle:

- **Detection System:** Flame sensors, often placed in the front and at corners, continuously scan for infrared light emitted by fire. Some systems also use cameras and thermal technology for enhanced detection.
- **Navigation & Control:** A microcontroller (Arduino) Processes input from the sensors. It decides the robot's movement moving forward to find the fire or turning left/right if the flame is detected off – centre.
- **Fire Suppression Mechanism:** Once the robot reaches the fire, it stops. A 5V relay module activates a water pump, while a servo motor moves the nozzle to spray water and cover the fire area,
- **Autonomous or Remote Operation:** The robot can be fully autonomous (running on programmed logic) or remote-controlled by an operator, ensuring safety by entering hazardous areas.

Operating Modes :-

Most advanced fire fighter robots are programmed with two distinct modes:

1. **Manual Mode:** You act as the "driver" via a Bluetooth app. You see the fire (either directly or via an onboard camera) and steer the robot to the exact spot.
2. **Automatic Mode:** The robot uses three sensors (Left, Center, Right) to scan the room.
 - a) If the Left sensor detects fire, the robot turns left.
 - b) If the Center sensor detects fire, it moves straight and activates the pump.
 - c) Once the sensor signal goes "LOW" (fire is out), the robot stops the pump.

1. Manual Mode: Human-in-the-Loop

In Manual Mode, the robot is a "drone." It does nothing unless it receives a specific instruction from an external controller (usually a smartphone or tablet).

- **Communication Flow:**
 - a) **The Interface:** You press a button on a Bluetooth Serial App.
 - b) **The Signal:** The phone sends a specific character (e.g., 'F' for Forward, 'S' for Spray) via the Bluetooth frequency (2.4 GHz).

- c) **The Reception:** The HC-05/06 Bluetooth Module receives this character and passes it to the microcontroller via UART communication (TX/RX pins).
- d) **The Execution:** The microcontroller runs a switch-case or if-else loop. If it sees 'F', it tells the Motor Driver to spin the wheels forward.
- **Best Use Case:** When the fire is in a complex environment with obstacles (furniture, stairs, or narrow gaps) that the robot's sensors might struggle to navigate autonomously.
- **Critical Weakness:** It relies on human reaction time and a stable Bluetooth connection. If the robot goes out of range (approx. 10 meters), it may "freeze" or continue its last command.

How to control HC-05 Bluetooth by Mobile Application :-

To control an HC-05 Bluetooth module via a smartphone app, you must first pair the module with your phone and then use a compatible application to send serial commands.

1. Initial Setup and Pairing :- Before using an app, you must link the HC-05 to your smartphone's Bluetooth settings:

- **Power the Module:** Connect the HC-05's VCC to 5V and GND to GND on your microcontroller (like an Arduino). A blinking LED on the module indicates it is ready to pair.
- **Enable Bluetooth:** On your phone, go to **Settings > Bluetooth** and scan for new devices.
- **Pair Device:** Select "HC-05" from the list. If prompted for a PIN, use the default password, which is usually **1234** or **0000**.
- **Check Permissions:** On newer Android versions, ensure the controller app has "Nearby Devices" or "Location" permissions enabled to scan for the module.

2. Recommended Controller Apps :- Depending on your project (e.g., a car, home automation, or simple testing), you can download specialized apps from the Google Play Store:

- **Arduino Bluetooth Controller:** A versatile app offering "Terminal Mode" for text commands and "Switch Mode" for simple On/Off control.
- **Bluetooth Terminal HC-05 :** Best for debugging; it allows you to send and receive raw ASCII or HEX data.
- **RC Bluetooth Controller(HC-05):** Specifically designed for controlling RC cars with directional pads.
- **Remote X Y:** Allows you to create a custom graphical user interface (buttons, sliders, joysticks) for your specific project.

3. Basic Operating Steps :- Once paired and the app is installed:

1. **Open the App:** Launch your chosen controller application.
2. **Connect:** Tap the **Connect** or **Bluetooth icon** within the app and select your paired "HC-05" from the list.
3. **Configure Commands:** In "Button" or "Controller" modes, you may need to map specific characters to buttons (e.g., setting a button to send '1' for ON and '0' for OFF).
4. **Send Data:** Tap the buttons or type in the terminal to send data wirelessly to your microcontroller

As you download and open the software, it will appear like an image & we can see in below figure.

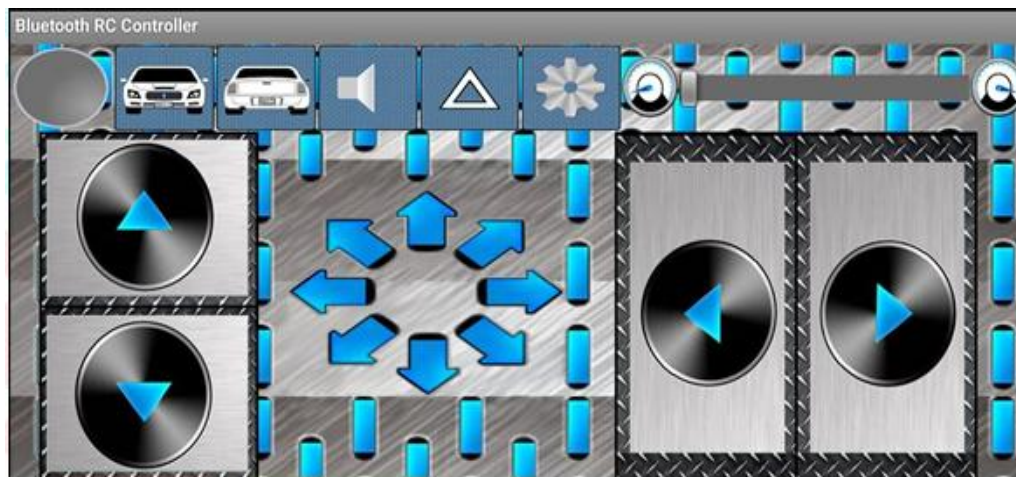


Fig 2. Robot Communicate With Android App

2. Automatic Mode: Sensor-Driven Intelligence

In Automatic Mode, the robot ignores Bluetooth signals and relies entirely on its Flame Sensor Array. This is where the robot acts as an "Autonomous Agent."

The Logic Tree (How it "Thinks")

The robot uses a "Search and Destroy" algorithm based on the placement of IR flame sensors (typically three sensors positioned at 45° angles):

- **State A (The Scan):** If no sensors detect IR light above a certain threshold, the robot may rotate in place or stay stationary to "scan" the room.
- **State B (Targeting):**
 - a) **Left Sensor High:** The robot triggers the right motor and stops the left motor, pivoting toward the heat source.
 - b) **Right Sensor High:** The robot triggers the left motor to pivot right.
- **State C (Engagement):** When the Center Sensor hits a maximum threshold (meaning the fire is directly in front and close), the robot stops all movement and triggers the Relay. The relay closes the circuit for the water pump, and the nozzle begins spraying.
- **State D (Confirmation):** The pump stays on until the Center Sensor's IR reading drops below the threshold, signaling the fire is extinguished.

The "Threshold" Concept :-

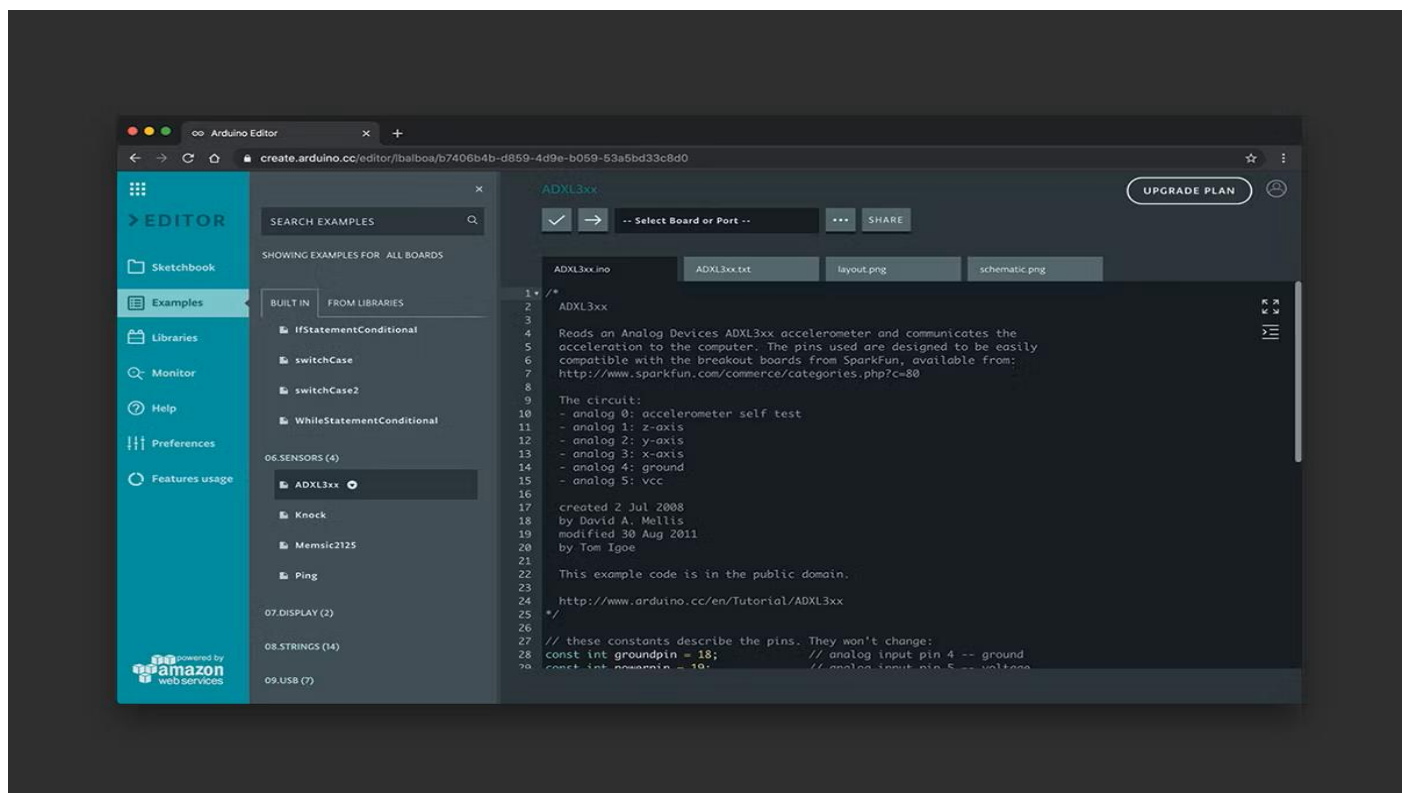
In Automatic Mode, the robot uses Analog-to-Digital Conversion (ADC). The flame sensor sends a voltage signal:

- **No Fire:** High voltage (e.g., 5V).
- **Fire Detected:** Low voltage (e.g., 0V - 1V). The code is usually written to trigger action only when the value drops below a specific number (the Threshold), which prevents the robot from being confused by sunlight or bright overhead lamps.

Software Required for Coding :-

- Arduino IDE is an open-source software that is mainly used for writing and compiling the Code into the Arduino Module.
- It is an official Arduino software, making code compilation too easy that even a common Person with no prior technical knowledge can get their feet wet with the learning process.
- It is easily available for operating systems like MAC, Windows, Linux and runs on the Java Platform that comes with inbuilt functions and commands that play a vital role for debugging, Editing and compiling the code in the environment.
- Each of them contains a microcontroller on the board that is actually programmed and accepts the information from on board is in the form of code.
- The main code, also known as a sketch, created on the IDE plat for generate a hex file which is Arduino Micro and many more then transferred and uploaded in the controller on the board the IDE environment mainly contains two basic parts: Editor and Compiler where former is Used for writing the required code and later is used for compiling and uploading the code into the given Arduino Module.
- This environment supports both C and C++ languages.
- 1.5.1 How to Download You can download the Software from Arduino main website. As I said Earlier, the software is for common operating systems like Linux, Windows, and MAX, so make sure you are downloading the correct software version that is easily Compatible with your operating system.
- If you aim to download Windows app version, make sure you have Windows 8.1 or Windows 10, as app version is not compatible with Windows 7 or older version of this operating system. The IDE environment is mainly distributed into three sections.
 - a) Menu Bar
 - b) Text Editor
 - c) Output Pan

As you download and open the IDE software, it will appear like an image & we can see in below figure.



Coding :-

```
#include <Wire.h>
#include <Servo.h>
#include <Adafruit_MLX90614.h>
#include <SoftwareSerial.h>

Adafruit_MLX90614 mlx;
Servo myservo;
SoftwareSerial bluetooth(13, 12); // RX, TX

#define Left 8
#define Right 9
#define Forward 10
#define LM1 2
#define LM2 3
#define RM1 4
#define RM2 5
#define pump 6

const int toggleSwitchPin = 7; // Pin for the toggle switch
bool manualControl = false;

void setup() {
  pinMode(Left, INPUT);
  pinMode(Right, INPUT);
  pinMode(Forward, INPUT);
  pinMode(LM1, OUTPUT);
  pinMode(LM2, OUTPUT);
  pinMode(RM1, OUTPUT);
  pinMode(RM2, OUTPUT);
  pinMode(pump, OUTPUT);

  pinMode(toggleSwitchPin, INPUT_PULLUP); // Use internal pull-up resistor
```

```
mlx.begin();
myservo.attach(11);
myservo.write(90);

bluetooth.begin(9600);
}

void sweepServo() {
  for (int pos = 50; pos <= 130; pos += 1) {
    myservo.write(pos);
    delay(10);
  }
  for (int pos = 130; pos >= 50; pos -= 1) {
    myservo.write(pos);
    delay(10);
  }
}

void put_off_fire()
{
  digitalWrite(LM1, LOW);
  digitalWrite(LM2, LOW);
  digitalWrite(RM1, LOW);
  digitalWrite(RM2, LOW);

  digitalWrite(pump, HIGH);
  {
    sweepServo(); // Sweep the servo while extinguishing the fire
  }
  digitalWrite(pump, LOW);
  myservo.write(90); // Center the servo
}

void automaticControl()
{

```

```

int leftSensor = digitalRead(Left);
int rightSensor = digitalRead(Right);
int forwardSensor = digitalRead(Forward);

if (leftSensor && rightSensor && forwardSensor) {
    digitalWrite(LM1, HIGH);
    digitalWrite(LM2, HIGH);
    digitalWrite(RM1, HIGH);
    digitalWrite(RM2, HIGH);
} else {
    // Stop and adjust movement based on sensor inputs
    digitalWrite(LM1, HIGH);
    digitalWrite(LM2, LOW);
    digitalWrite(RM1, HIGH);
    digitalWrite(RM2, LOW);

    if (leftSensor == LOW) {
        // Turn left
        digitalWrite(LM1, HIGH);
        digitalWrite(LM2, LOW);
        digitalWrite(RM1, HIGH);
        digitalWrite(RM2, HIGH);
    }

    if (rightSensor == LOW) {
        // Turn right
        digitalWrite(LM1, HIGH);
        digitalWrite(LM2, HIGH);
        digitalWrite(RM1, HIGH);
        digitalWrite(RM2, LOW);
    }
}

void manualControlViaBluetooth() {

```



```
if (bluetooth.available() > 0) {  
    char command = bluetooth.read();  
    handleBluetoothCommand(command);  
}  
}  
  
void handleBluetoothCommand(char command) {  
    switch (command) {  
        case 'F':  
            // Move forward  
            digitalWrite(LM1, HIGH);  
            digitalWrite(LM2, LOW);  
            digitalWrite(RM1, HIGH);  
            digitalWrite(RM2, LOW);  
            break;  
        case 'B':  
            // Move backward  
            digitalWrite(LM1, LOW);  
            digitalWrite(LM2, HIGH);  
            digitalWrite(RM1, LOW);  
            digitalWrite(RM2, HIGH);  
            break;  
        case 'L':  
            // Turn left  
            digitalWrite(LM1, HIGH);  
            digitalWrite(LM2, LOW);  
            digitalWrite(RM1, LOW);  
            digitalWrite(RM2, HIGH);  
            break;  
        case 'R':  
            // Turn right  
            digitalWrite(LM1, LOW);  
            digitalWrite(LM2, HIGH);  
            digitalWrite(RM1, HIGH);
```

```

    digitalWrite(RM2, LOW);
    break;
case 'X':
    put_off_fire();
    break;
case 'S':
    // Stop
    digitalWrite(LM1, HIGH);
    digitalWrite(LM2, HIGH);
    digitalWrite(RM1, HIGH);
    digitalWrite(RM2, HIGH);
    break;
default:
    break;
}
}

void loop() {
    // Read the state of the toggle switch
    bool switchState = digitalRead(toggleSwitchPin);

    if (!switchState) {
        // Automatic control mode
        automaticControl(); // Read temperature only in automatic mode
        double temp = mlx.readObjectTempC();

        if (temp > 37.0) {
            put_off_fire();
        }
    } else {
        // Manual control mode
        manualControlViaBluetooth();
    }
}
}

```

Advantages :-

Robotics offers significant advantages including increased productivity, high-precision accuracy, enhanced workplace safety, 24/7 operational capability, and long-term cost savings. These machines efficiently handle dangerous or repetitive tasks, leading to better quality, reduced waste, and improved production speed, ultimately strengthening competitiveness.

1. **Increased Productivity and Efficiency** Robots can work continuously without breaks, fatigue, or vacations, producing a higher volume of work in a shorter time compared to human labor.
2. **High Precision and Consistency** Robots are designed for high-level accuracy, performing repetitive tasks with high precision. This minimizes human error, resulting in superior product quality and reduced defects.
3. **Improved Safety in Hazardous Environments** Robots can take over dangerous jobs, such as handling toxic materials, working in extreme temperatures, or operating in confined spaces, thus reducing workplace accidents and human injuries.
4. **Reduced Operational Costs** although initial investment is high, robots save money over time by reducing labor costs, cutting down on material waste, and reducing the need for redoing work.
5. **Increased Flexibility and Scalability** Robots can be reprogrammed to adapt to new tasks, providing versatility in manufacturing. They allow for seamless, quick changes to production lines to meet changing demands.

Disadvantages :-

Robotics presents significant drawbacks, including high initial implementation costs, potential for widespread job displacement, and limited adaptability to new tasks. Other major disadvantages are the need for specialized maintenance, lack of human-like creativity or empathy, and risks of downtime due to technical failures.

Here are five key disadvantages of robotics:

- **High Implementation Costs:** Purchasing, installing, and programming robotic systems requires substantial capital, which can be prohibitive for smaller businesses.
- **Job Displacement:** Robots often replace human workers in manufacturing and service sectors, leading to unemployment and the need for workforce retraining.
- **Limited Flexibility and Adaptability:** Robots are typically designed for specific, repetitive tasks. They struggle to adapt to unexpected changes or new situations without extensive reprogramming.
- **High Maintenance and Repair Needs:** Robots require specialized, skilled technicians for regular maintenance and repairs. A technical glitch can cause significant downtime.
- **Lack of Human Intelligence and Emotion:** Robots cannot understand complex emotions, make ethical judgments, or think creatively. Their reliance on human programming means they cannot operate outside their instructions.

Project Cost

Sr. No.	Component	Price	Quantity	Amount(Rs)
1.	Arduino Uno	250	1	250
2.	Battery charging module	90	1	90
3.	Bluetooth Module	300	1	300
4.	BO Moter	60	4	240
5.	Buzzer	50	1	50
6.	Cable Tie	2	20	40
7.	Glue Gun	120	1	120
8.	Glue Sticks	10	5	50
9.	IR Flame Sensor	100	3	300
10.	LED	5	5	25
11.	Lithium Battery	80	3	240
12.	Moter driver	150	1	150
13.	Multi-meter	180	1	180
14.	Nozzle	40	1	40
15.	PCB	40	1	40
16.	Ply wood	100	1	100
17.	Resistance	2	5	10
18.	Scale	10	1	10
19.	Screwdriver	50	1	50
20.	Servo Moter	250	1	250
21.	Single Channel Relay	60	1	60
22.	Soldering Iron	120	1	120
23.	Soldering Wire & Wick	70	1,1	70
24.	Switch	20	1	20
25.	Tweezer	20	1	20
26.	Type C Charger	250	1	250
27.	Water Pipe	40	1m	40
28.	Water Pump	80	1	80
29.	Water storage tank	50	1	50
30.	Wheels	40	4	160
31.	Wire Stepper	50	1	50
32.	Wooden cutter	10	1	10
33.	Other (Reports, Travelling, Brack-fast & some other Components)	2500	-	2500
			Total	5965

Future Scope of Bluetooth & Automatic Controlled Firefighter :-

The future scope of Bluetooth controlled firefighter robots is promising, with ongoing developments and innovative in the field. These robots are expected to play a crucial role in enhancing fire safety and response capabilities in various environments. Here are some potential future developments:

- **Advanced Sensing Capabilities:** Integration of gas sensors, thermal cameras, and smoke detectors to enhance early fire detection and improve accuracy.
- **AI and Machine Learning:** Implementing machine learning algorithms to enable smart decision-making based on sensor patterns and environmental data.
- **Real-time Video Surveillance:** Adding a camera module for live video streaming to allow remote monitoring of fire incidents and robot movement.
- **GPS and Navigation:** Incorporating GPS for precise location tracking and enabling the robot to operate in outdoor or large-scale environments like forests or farms
- **Obstacle Avoidance System:** Using ultrasonic or infrared sensors for intelligent path planning and obstacle detection to improve autonomous navigation.
- **Solar Power Integration:** Exploring solar energy options to reduce dependency on batteries and enhance energy efficiency.
- **Multi-Robot Coordination:** Developing a network of robots can work together in large or complex fire zones for faster and more effective fire suppression.

APPLICATION :-

- It is used to extinguish fire where possibility of explosion is more.
- It used as home security application.
- It is used in server rooms in offices.
- It is useful in disaster area monitoring and rescue.

RESULT

The IoT-Based Fire Fighter Robot was successfully designed, developed, and tested for fire detection and extinguishing operations. The system demonstrated reliable flame detection using infrared sensors, accurately identifying fire sources within a 60-degree field of view and up to a distance of 15 cm autonomously. The robot's response time from detection to water spray activation averaged 1.5 seconds, ensuring rapid suppression in simulated fire scenarios.

The servo motor-enabled nozzle system effectively directed the water stream towards the flame, achieving high targeting precision. During remote operation via Bluetooth, the robot maintained stable wireless communication with a smartphone up to a range of 10 meters, allowing full manual control of movement and fire suppression functions. The DC gear motors, controlled through the L298N driver, provided smooth and controlled mobility across flat surfaces. The water pump, activated by the relay module, delivered a steady flow rate suitable for small-scale fire suppression.

The system's energy efficiency was validated through continuous operation for up to 40 minutes on a fully charged Li-ion battery, with no significant voltage drop affecting system performance. The modular software design enabled seamless integration of hardware components, while the use of non-blocking code structures ensured responsive real-time behaviour. Overall, the robot effectively demonstrated its capability as a prototype firefighting tool, reducing the need for human intervention in dangerous environments.

Real Picture :-



CONCLUSION

In conclusion, The IoT-Based Fire Fighter Robot effectively demonstrates the integration of autonomous fire detection, real-time decision-making, and remote-control capabilities for small-scale firefighting applications. The system's core functionality reliable flame detection using infrared sensors, efficient navigation via DC gear motors, and precise fire suppression through a servo-controlled nozzle shows strong potential for deployment in hazardous indoor environments. The Bluetooth-enabled manual control system provides operators with situational flexibility. Experimental results validate the robot's rapid response time, stable communication, and efficient power utilization through the Li-ion battery. the IoT-Based Fire Fighter Robot offers a promising, energy-efficient solution for reducing human risk during firefighting operations, paving the way for intelligent, autonomous firefighting systems in industrial and residential applications.

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